Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/23/2024
PI Name:	Panontin, Tina Ph.D.		
Project Title:	Human Systems Integration Architecture: Capturing Crew Responses on Artemis Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2022	**End Date:**	10/01/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Shu-Chieh (San Jose State University Research Foundation Inc) McTigue, Kaitlin M.S. (NASA Ames Research Center) Vera, Alonso Ph.D. (NASA Ames Research Center) Parisi, Megan (NASA Ames Research Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Distance from Earth will reduce ground support for the execution of complex nominal and off-nominal safety-critical operations during future deep space exploration missions. The risk of a loss-of-mission outcome for Artemis lunar surface operations is greater than 10% for a 30-day mission. This is characterized as a Human Systems Integration Architecture (HSIA) risk because mitigation depends on identifying onboard capability requirements and crew roles and responsibilities necessary to enable effective and efficient crew responses to the aforementioned situations. Previous work focused on deriving onboard crew capability needs by observing and analyzing anomaly response processes associated with incidents on the International Space Station (ISS). Though yielding an extremely rich set of data, processes revealed from ISS incidents were predominantly ground-driven, reflecting current operation models. These therefore offer limited insight into how a small flight crew will have to execute similar anomaly responses on their own or coordinate with the ground at significant communication delays. The purpose of the present work is to develop an approach to systematically capture and assess crew operational performance during early Artemis missions, particularly pertaining to safety-critical operations such as anomaly resolution and procedure execution. Being the first human spaceflight beyond low Earth orbit (LEO) since Apollo, early Artemis missions, with their increasingly complex mission designs and durations, provide a rare and unique opportunity to serve as a testbed for Mars missions. The team surveyed available and planned crew-related data sources for Artemis missions, identified data needed to reconstruct anomaly resolution and procedure execution processes using prior work analyzing historical ISS and Apollo anomaly responses, and outlined an approach to systematically capture and assess crew operational performance. The present work is most closely related to Gaps HSIA-401 and HSIA-501, but also has general implications for other HSIA Gaps. The work addresses the crew performance issues raised in those gaps by supporting crew capability needs.
Rationale for HRP Directed Research:	This proposal qualifies for a directed study due to the urgent, time-sensitive need to develop HSIA risk characterization research for Artemis missions and beyond. The Artemis II mission scheduled for August 2023 results in a compressed timeline to prepare research, determine data collection protocols, and collect and analyze data. Timely data analysis can inform future Artemis missions, and results can be used as baseline data for future Mars related mission requirements. Additionally, the directed nature of this study allows the HSIA Discipline Scientist to leverage previously collected data that may inform solutions.
Research Impact/Earth Benefits:	Our research program and associated deliverables will contribute to reducing the Earth Independent Human-Systems Operations (EIHSO) risk, formerly known as Human-Systems Integration Architecture (HSIA) risk. They will also inform gaps in knowledge relevant to maintaining crew health and performance on future long duration exploration missions beyond low-Earth orbit (LEO).
Task Progress:	Final presentation of the project, entitled “Human Systems Integration Architecture: Capturing Crew Responses on Artemis Missions,” was delivered to the Human Factors and Behavioral Performance Element on November 29, 2023.
Bibliography Type:	Description: (Last Updated: 01/29/2025)